MARK HOSPITALS

HOSPITAL MANAGEMENT SYSTEM

PROJECT REPORT

A role-oriented dashboard for hospital operations

SUBMISSION DETAILS	INFORMATION
Course	24CS810 - Modern Front-End Frameworks and SPA Development
Student Name	A. Tejaswini
Registration Number	241FA04730
Section	09
Faculty	_____

MARK Hospitals HMS

Prepared with reference to the supplied project-report format

1. Project Overview

The MARK Hospitals Hospital Management System (HMS) is a browser-based demonstration dashboard for common hospital workflows. It presents a single interface with role-oriented screens for administration, clinical care, reception, pharmacy, finance, and patients. The current workspace combines a React/Vite application shell with a standalone HTML dashboard containing the interface, sample records, and client-side interactions.

Aim

To develop a user-friendly hospital management dashboard that demonstrates how patient information, appointments, staff views, pharmacy operations, billing, and reports can be organized in one web application.

Objectives

- ? Provide a dashboard view of hospital activity and operational summaries.
- ? Demonstrate patient registration, admission, and patient-list workflows.
- ? Support appointment booking and appointment status displays.
- ? Present role-specific navigation and screens for six hospital roles.
- ? Show pharmacy inventory, prescriptions, billing, finance, lab, and reporting views.
- ? Provide a maintainable starting point for future API, database, and authentication integration.

Project Scope

This version is a front-end demonstration using in-memory sample data. It is suitable for academic demonstration and interface prototyping, not for clinical use or storage of real patient information.

2. Software and Hardware Requirements

Category	Requirement
Operating system	Windows 10/11 (or another OS supported by Node.js)
Runtime	Node.js 18 or newer; npm
Frontend	React 19, React DOM 19, JavaScript, HTML5, CSS3
Build / development server	Vite 6 with the Vite React plugin
Code editor	Visual Studio Code or another JavaScript editor
Browser	A current version of Chrome, Edge, Firefox, or equivalent
Storage / backend	No backend or database is included; dashboard records are in-memory demo data
Hardware	A standard desktop or laptop capable of running a modern browser and Node.js

Run the Project

- ? Install dependencies with npm install.
- ? Start the development server with npm run dev.
- ? Open the local address printed by Vite in a browser.
- ? Create a production build with npm run build.

The Continue button enters the demonstration dashboard as Administrator; it does not authenticate a real hospital account.

3. Project Structure

```
hms/
|-- index.html
|-- package.json
|-- README.md
|-- public/
|   `-- hospital-dashboard.html
|-- src/
|   |-- App.jsx
|   |-- main.jsx
|   `-- styles.css
|-- node_modules/            (installed dependencies)
`-- dist/                    (generated by npm run build)
```

The source files are intentionally compact in the current version. The hospital UI and its demo data are in public/hospital-dashboard.html; the React application mounts that page in an iframe. node_modules and dist are generated folders and should not be edited as source files.

Technology and Architecture

- ? index.html provides the root element and loads the React entry point.
- ? src/main.jsx mounts the React application and imports the outer app styles.
- ? src/App.jsx embeds /hospital-dashboard.html and presents a fallback if the frame fails to load.
- ? src/styles.css controls the full-screen dashboard host and iframe layout.
- ? public/hospital-dashboard.html contains the visual interface, embedded CSS, role navigation, sample data, and browser-side event handlers.
- ? Vite serves the development app and bundles the React shell for production.

Data Flow

Browser -> Vite-served React entry point -> App component -> hospital-dashboard.html -> client-side demo data and DOM updates. Form actions update JavaScript arrays in the current page session. There is no REST API, server-side validation, database, or durable save operation in the present implementation.

4. Functional Modules

Role / Area	Representative screens and functions
Administrator	Dashboard; patient management; doctors and staff; appointments; billing and finance; pharmacy; reports.
Doctor	Doctor dashboard, patient list, appointments, prescriptions, and lab results.
Patient	Personal dashboard, appointment history, lab reports, and bills.
Receptionist	Reception dashboard, appointments, patient registration, and patient list.
Pharmacist	Pharmacy dashboard, inventory, prescriptions, and supplier orders.
Finance	Finance dashboard, invoices, revenue reports, and insurance claims.
Shared interface	Sidebar navigation, status labels, tables, dialogs, notifications, and live date/time display.

Main Demonstration Workflows

- ? Patient admission: enter the required name and diagnosis fields, then add a patient record to the in-memory data collection.

- Appointment booking: provide a patient name and date, choose appointment details, and append a scheduled appointment.
- Role navigation: open the dashboard sections available to the selected demonstration role.
- Pharmacy and finance review: inspect sample stock, prescription, invoice, revenue, and claim data.

5. Implementation Excerpts

React host component (src/App.jsx):

```
return (
  <main className="dashboard-host">
    <iframe
      className="dashboard-frame"
      src="/hospital-dashboard.html"
      title="MARK Hospitals management dashboard"
      onError={() => setLoadError(true)}
    />
  </main>
);
```

Patient admission validation and demo-record creation (public/hospital-dashboard.html):

```
function admitPatient() {
  const fname = document.getElementById('ap-fname').value.trim();
  const lname = document.getElementById('ap-lname').value.trim();
  const diag = document.getElementById('ap-diag').value.trim();
  if (!fname || !lname || !diag) {
    toast('Please fill in required fields');
    return;
  }
  DB._patientCounter++;
  DB.patients.push({ id, fname, lname, diag, admitted: today() });
  refreshCurrentPage();
}
```

The excerpt is abbreviated for the report. The live function also reads the other admission fields, assigns an ID and display values, closes and clears the dialog, and shows a confirmation message.

6. Verification and Limitations

Verification performed for this report: the Vite production build was run from the project root. Manual feature checks should also confirm that the dashboard loads, navigation switches views, required-field validation appears, patient admission and appointment forms update their tables, and the layout remains usable at narrow viewport widths.

Known Limitations

- The application is a demo, not a production hospital information system.
- The dashboard currently opens in Administrator demo mode; there is no real login, identity verification, or role authorization.
- Records are in browser memory and are lost when the page is reloaded; no API or database is configured.
- Sample data and interface actions do not represent validated medical records, clinical decision support, or production billing.
- No production-grade audit trail, access controls, backup, encryption, or privacy safeguards are implemented.

Privacy and safety: do not enter real patient information or use this demonstration to make clinical or financial decisions.

Future Enhancements

- Separate UI modules into reusable React components and add automated tests.

- ? Add a validated backend API with persistent database storage and role-based authorization.
- ? Use secure authentication, audit logging, backups, and appropriate privacy controls before handling sensitive records.
- ? Add real workflows for appointments, prescriptions, inventory updates, billing, and report exports with server-side validation.

Conclusion

MARK Hospitals HMS demonstrates a broad hospital operations interface and role-oriented workflows using React, Vite, and browser-side JavaScript. The project is a useful academic front-end prototype. Persistent data, secure identity management, and clinical-grade safeguards should be implemented and independently validated before any real-world use.

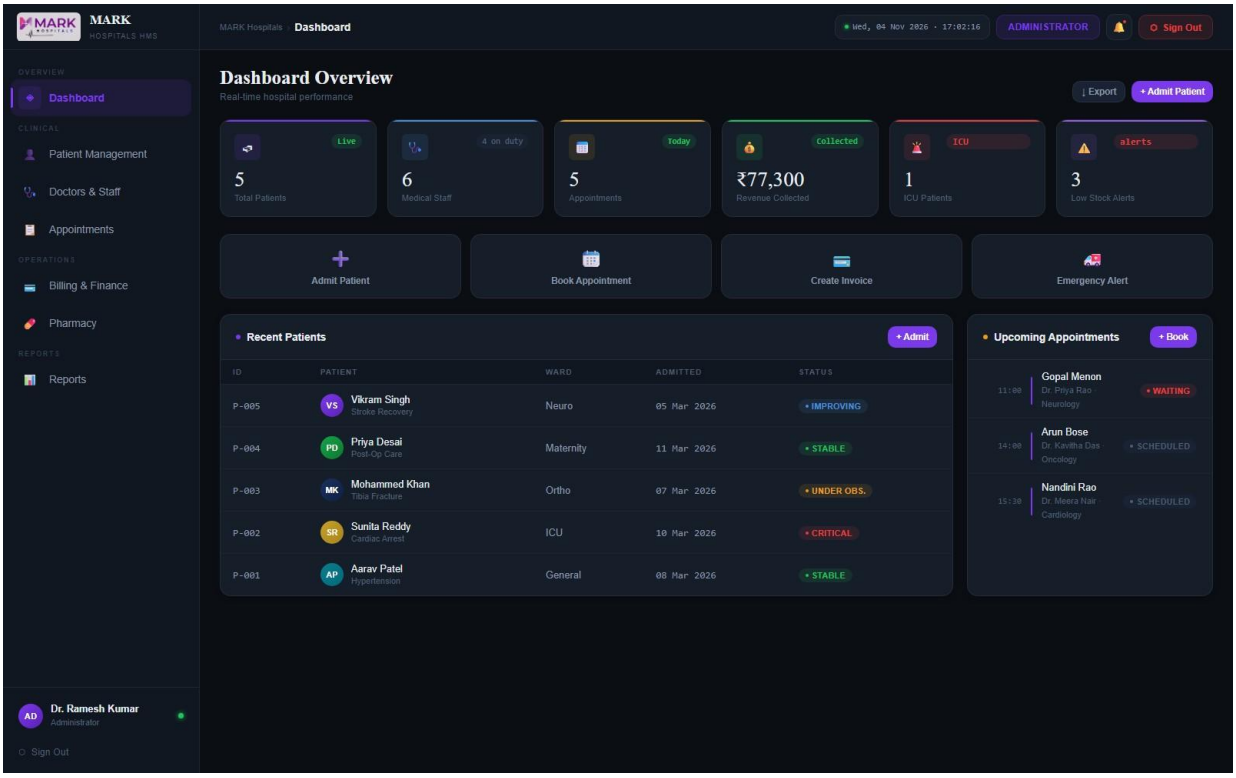
Prepared by A. Tejaswini | Registration No. 241FA04730 | Section 09

7. HMS Interface Screenshot Gallery

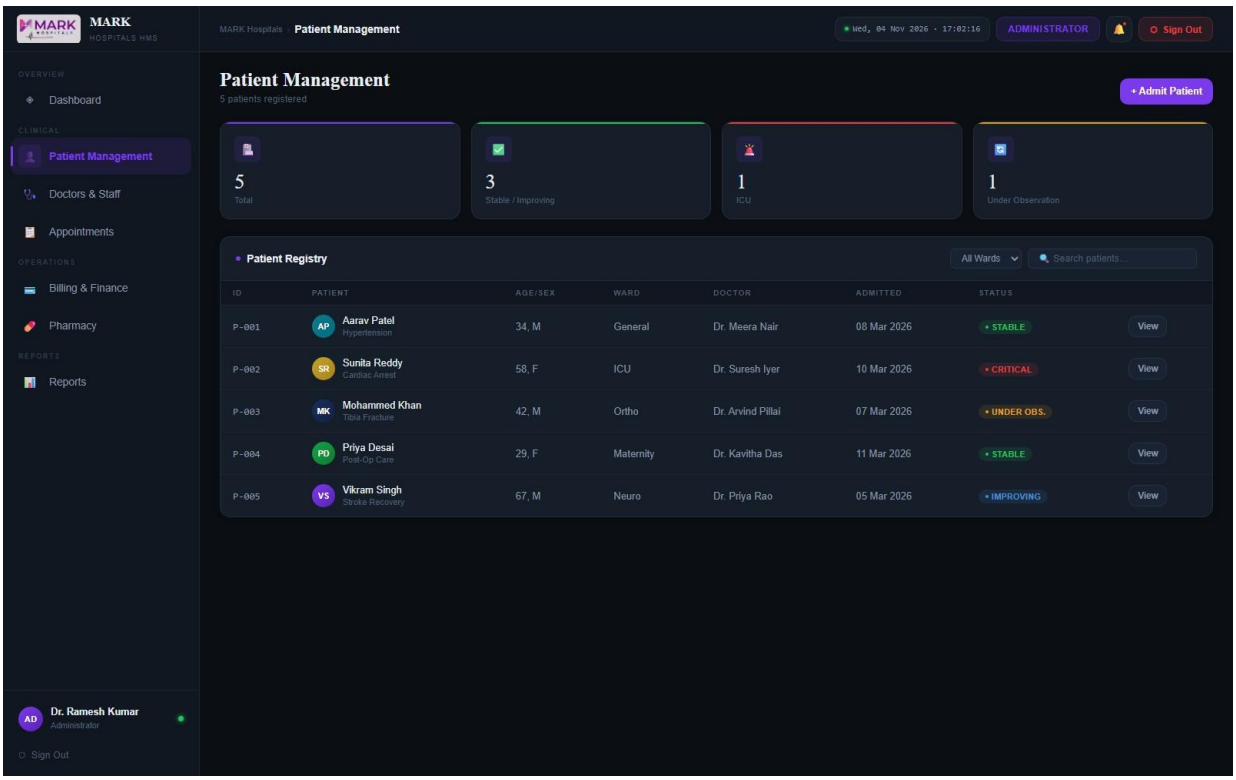
The following screenshots show the generated MARK Hospitals HMS interface. Screens are grouped by dashboard role. The displayed records are sample demonstration data.

Administrator

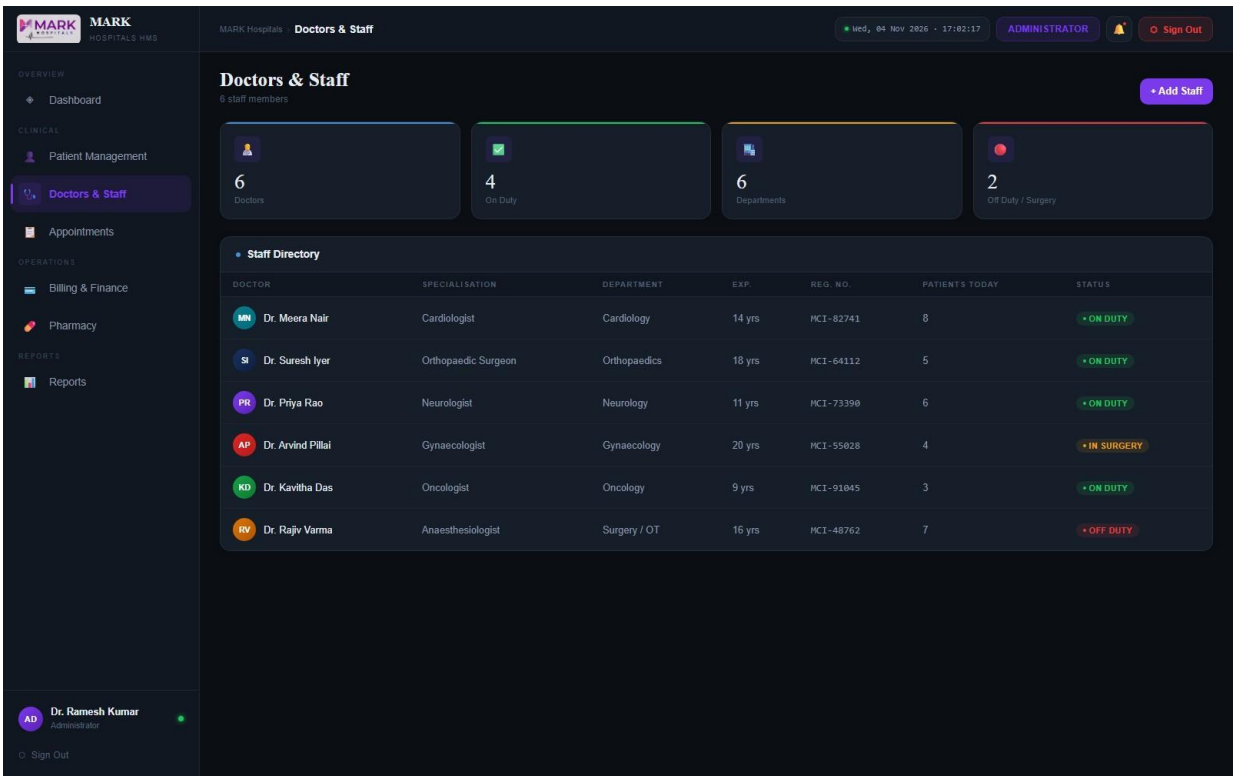
Dashboard



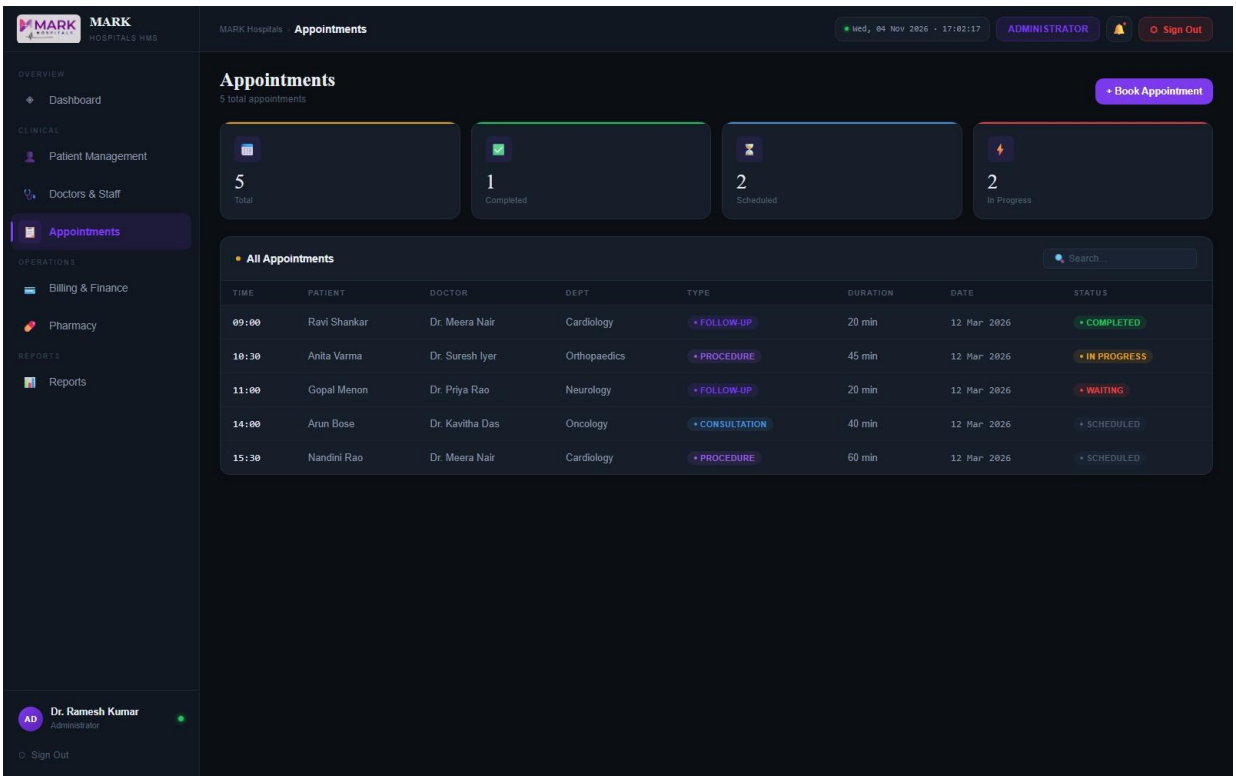
Patient Management



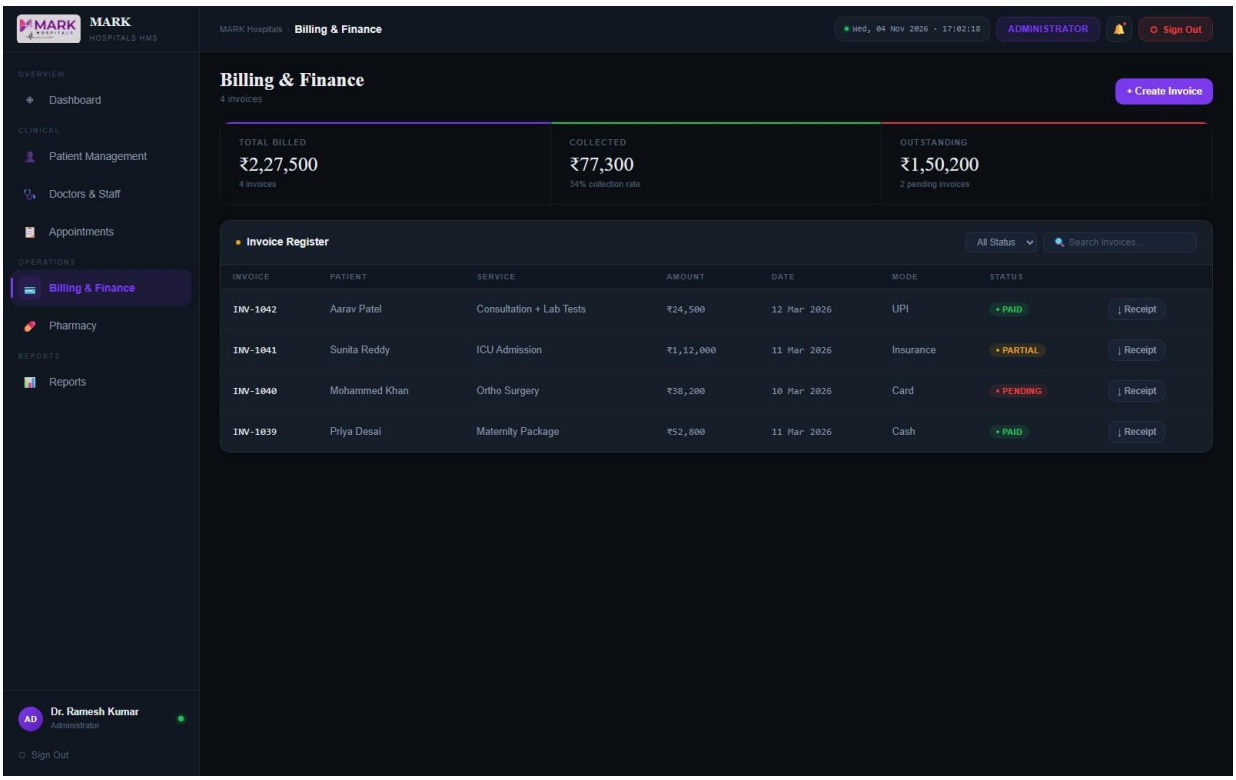
Doctors & Staff



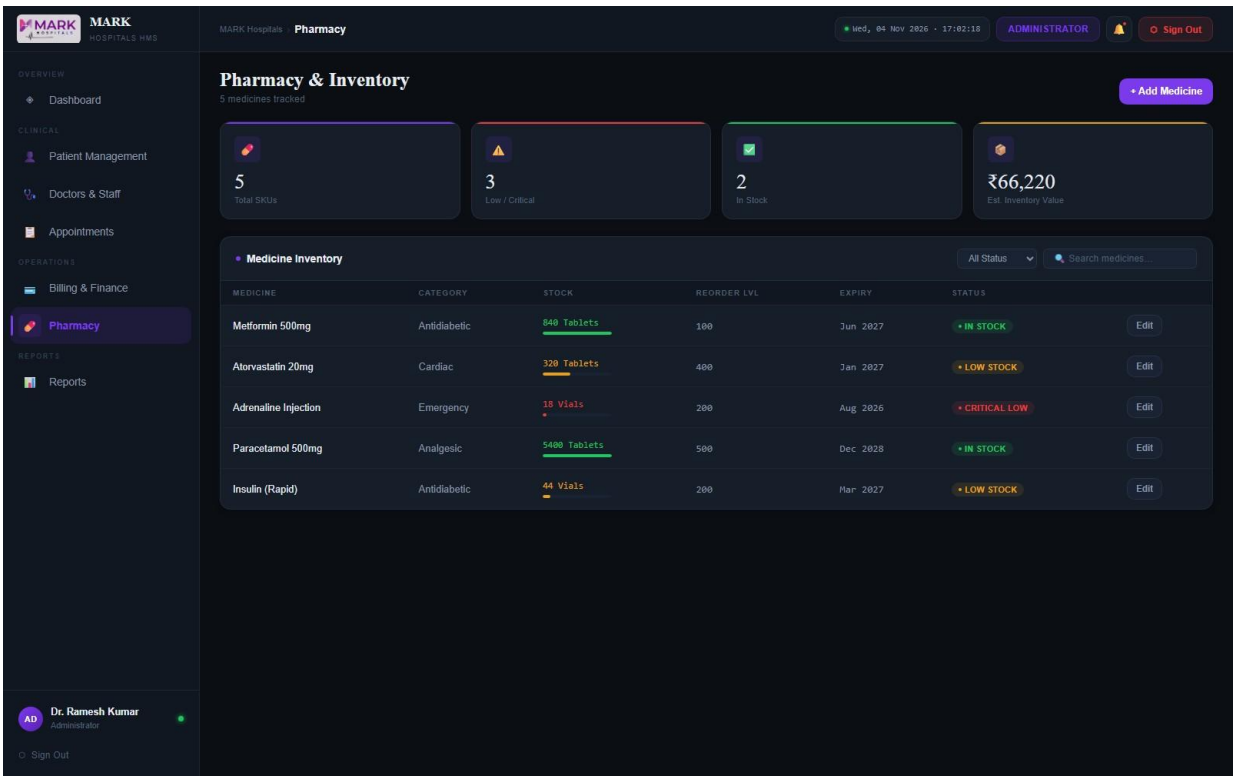
Appointments



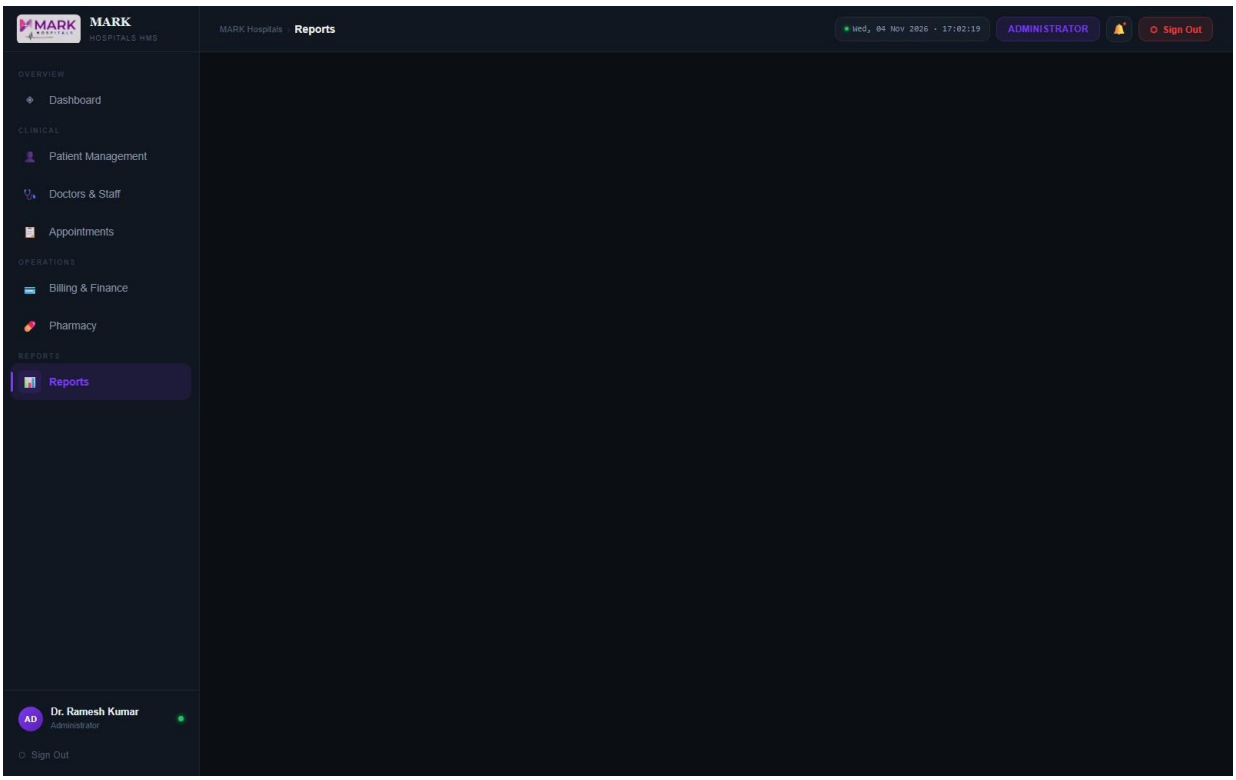
Billing & Finance



Pharmacy

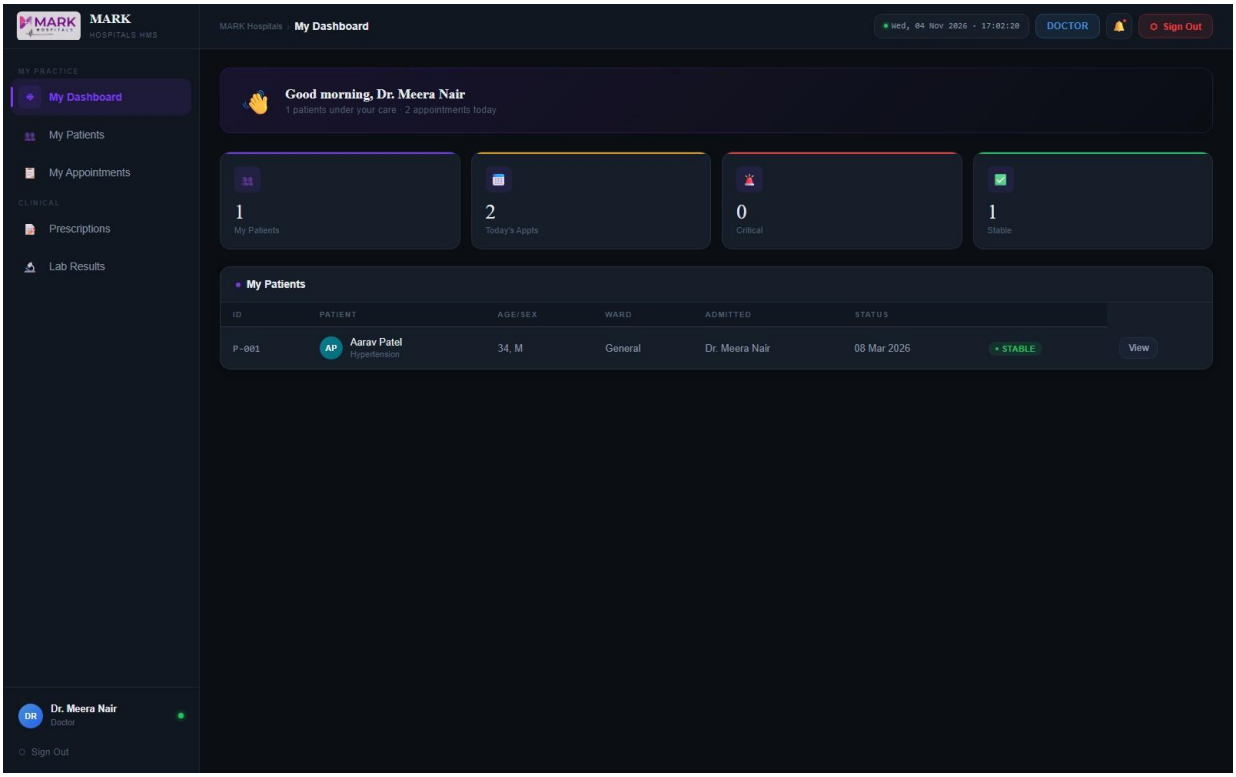

MARK
HOSPITALS HMS
MARK Hospitals › Pharmacy
Wed, 04 Nov 2026 · 17:02:18
ADMINISTRATOR
Sign Out
OVERVIEW
Dashboard
CLINICAL
Patient Management
Doctors & Staff
Appointments
OPERATIONS
Billing & Finance
Pharmacy
REPORTS
Reports
Pharmacy & Inventory
5 medicines tracked
+ Add Medicine
5
Total SKUs
3
Low / Critical
2
In Stock
₹66,220
Est. Inventory Value
Medicine Inventory
All Status
Search medicines...
MEDICINE
CATEGORY
STOCK
REORDER LVL
EXPIRY
STATUS
Metformin 500mg
Antidiabetic
840 Tablets
100
Jun 2027
IN STOCK
Edit
Atorvastatin 20mg
Cardiac
320 Tablets
400
Jan 2027
LOW STOCK
Edit
Adrenaline Injection
Emergency
18 Vials
200
Aug 2026
CRITICAL LOW
Edit
Paracetamol 500mg
Analgesic
5400 Tablets
500
Dec 2028
IN STOCK
Edit
Insulin (Rapid)
Antidiabetic
44 Vials
200
Mar 2027
LOW STOCK
Edit
AD
Dr. Ramesh Kumar
Administrator
Sign Out

Reports

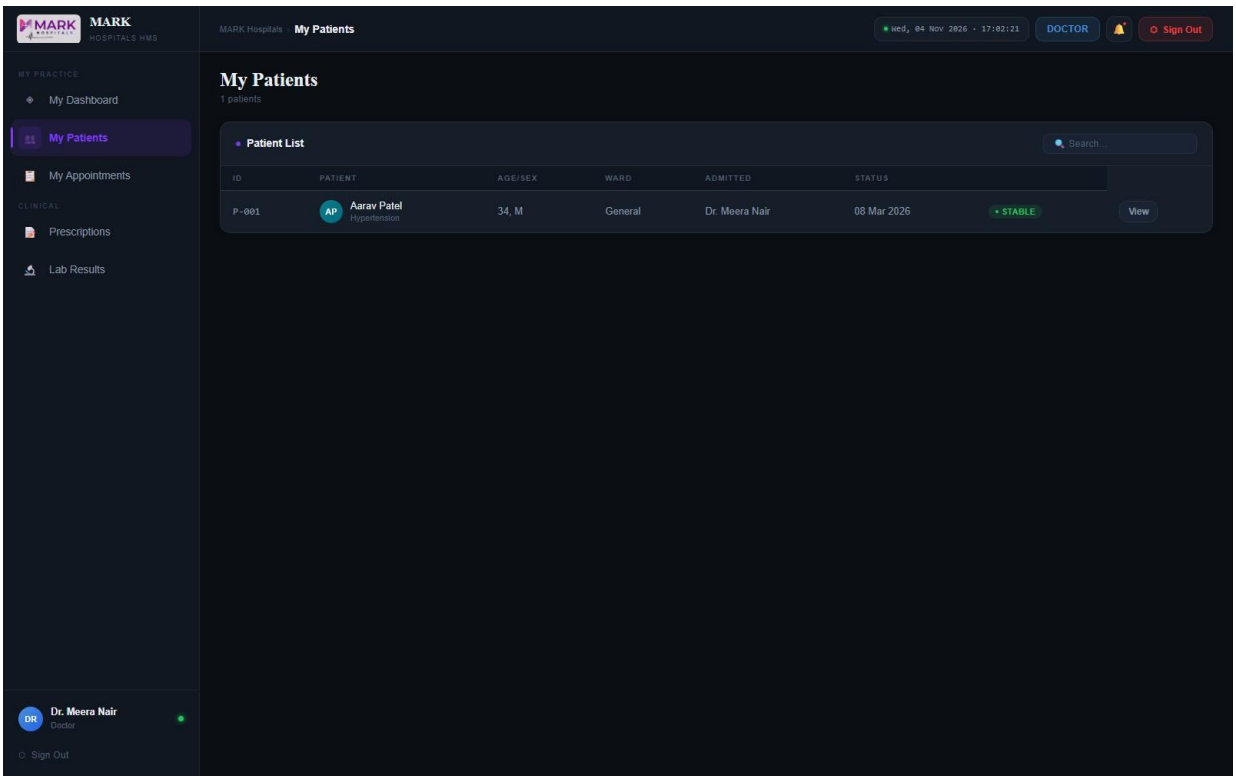

MARK
HOSPITALS HMS
MARK Hospitals › Reports
Wed, 04 Nov 2026 · 17:02:19
ADMINISTRATOR
Sign Out
OVERVIEW
Dashboard
CLINICAL
Patient Management
Doctors & Staff
Appointments
OPERATIONS
Billing & Finance
Pharmacy
REPORTS
Reports
AD
Dr. Ramesh Kumar
Administrator
Sign Out

Doctor

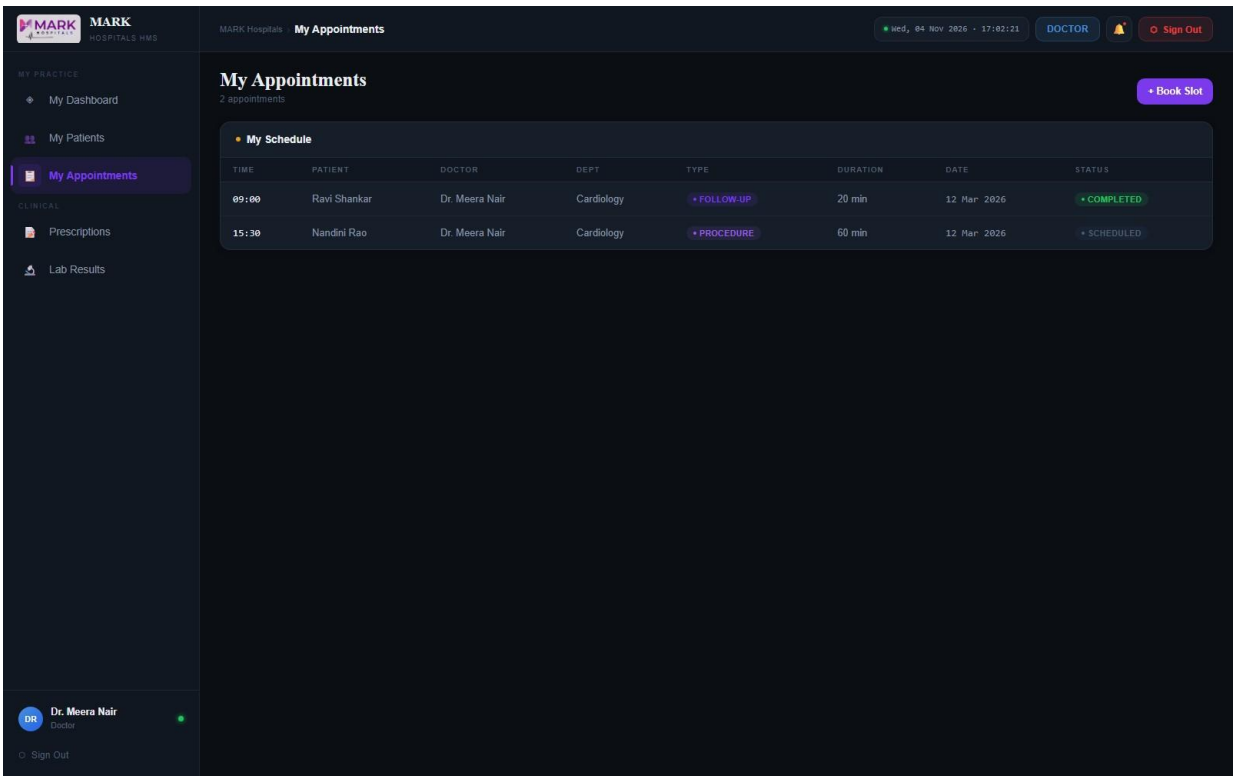
My Dashboard



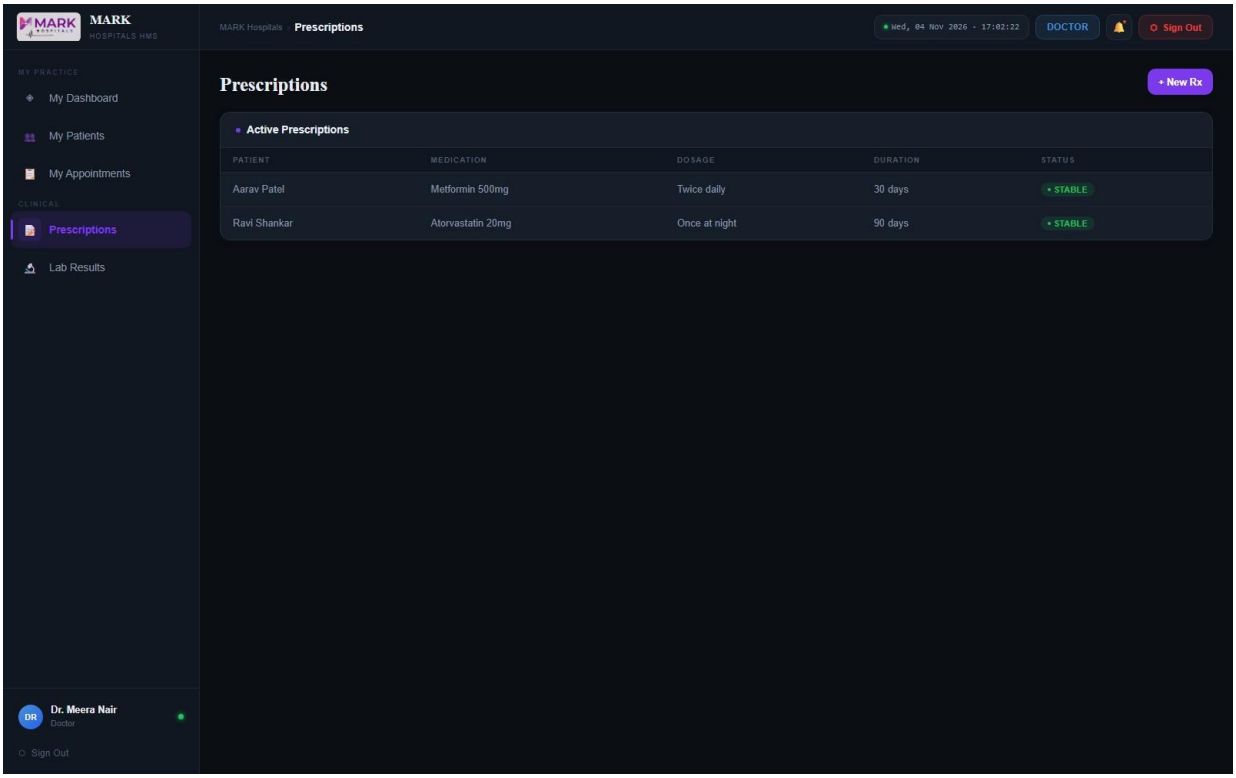
My Patients


MARK
HOSPITALS HMS
MARK Hospitals
My Patients
Wed, 04 Nov 2026 · 17:02:21
DOCTOR
Sign Out
MY PRACTICE
My Dashboard
My Patients
My Appointments
CLINICAL
Prescriptions
Lab Results
My Patients
1 patients
Patient List
Search...
ID
PATIENT
AGE/SEX
WARD
ADMITTED
STATUS
P-001
AP
Aarav Patel
Hypertension
34, M
General
Dr. Meera Nair
08 Mar 2026
STABLE
View
DR
Dr. Meera Nair
Doctor
Sign Out

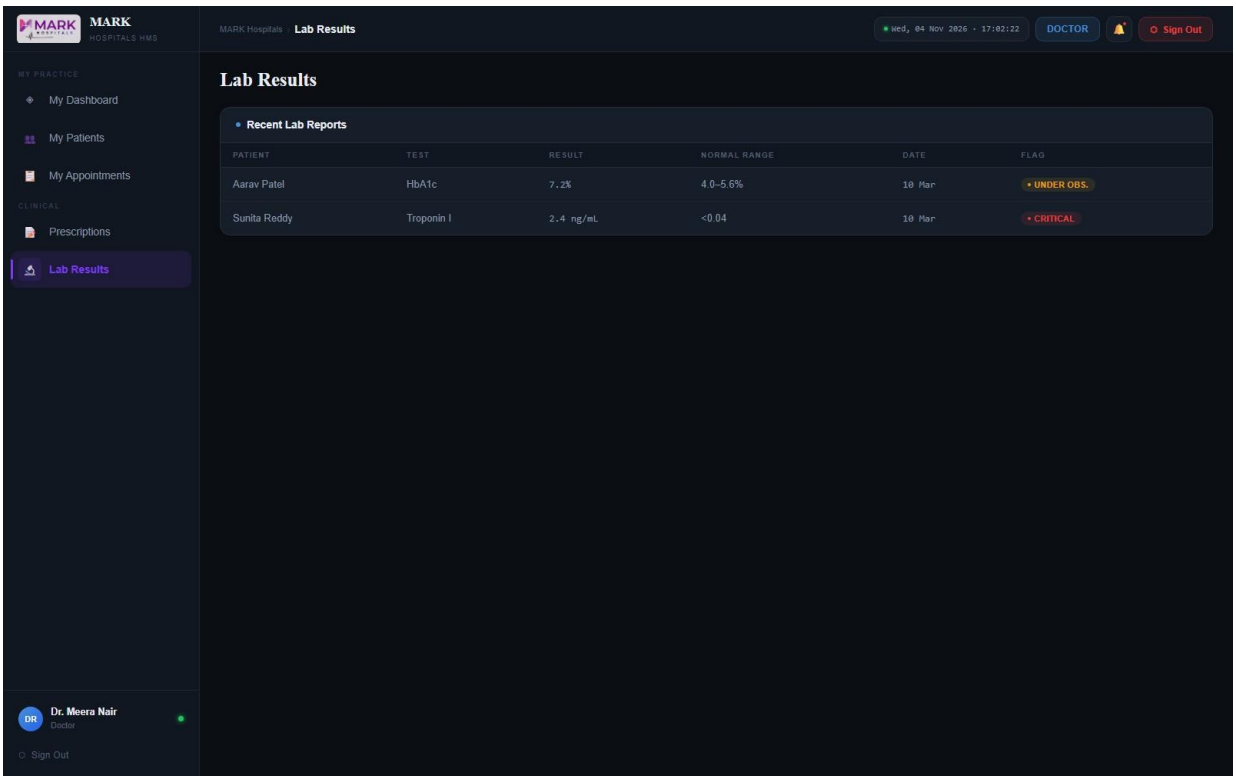
My Appointments



Prescriptions

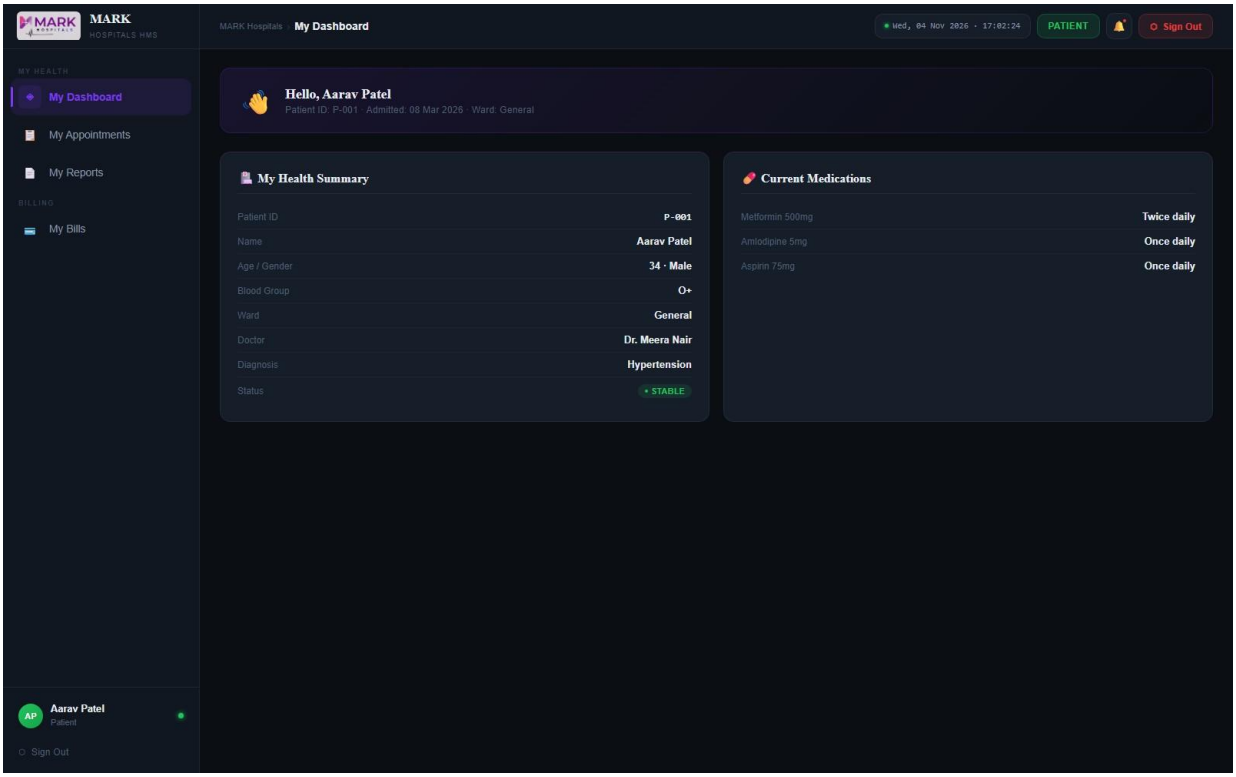


Lab Results

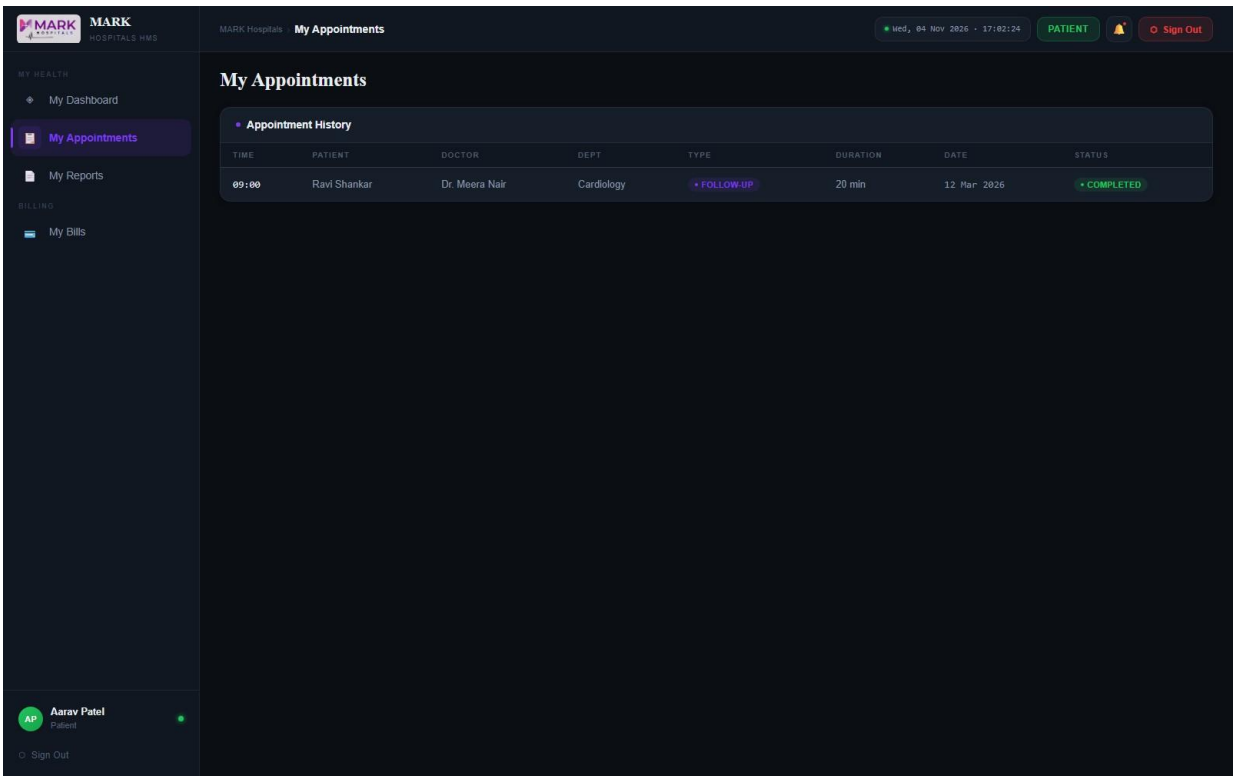

MARK
HOSPITALS HMS
MY PRACTICE
My Dashboard
My Patients
My Appointments
CLINICAL
Prescriptions
Lab Results
Dr. Meera Nair
Doctor
Sign Out
MARK Hospitals
Lab Results
Wed, 04 Nov 2026 · 17:02:22
DOCTOR
Sign Out
Lab Results
Recent Lab Reports
PATIENT
TEST
RESULT
NORMAL RANGE
DATE
FLAG
Aarav Patel
HbA1c
7.2%
4.0–5.6%
10 Mar
UNDER OBS.
Sunita Reddy
Troponin I
2.4 ng/mL
<0.04
10 Mar
CRITICAL

Patient

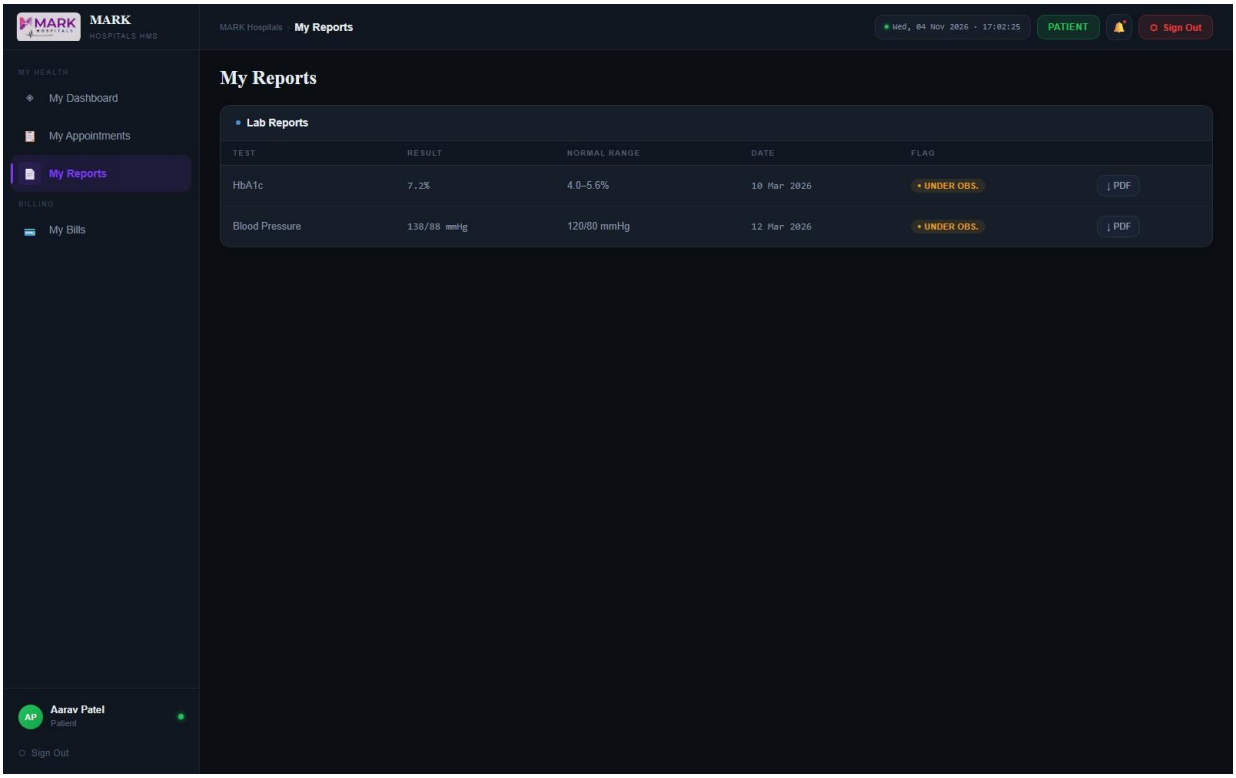
My Dashboard



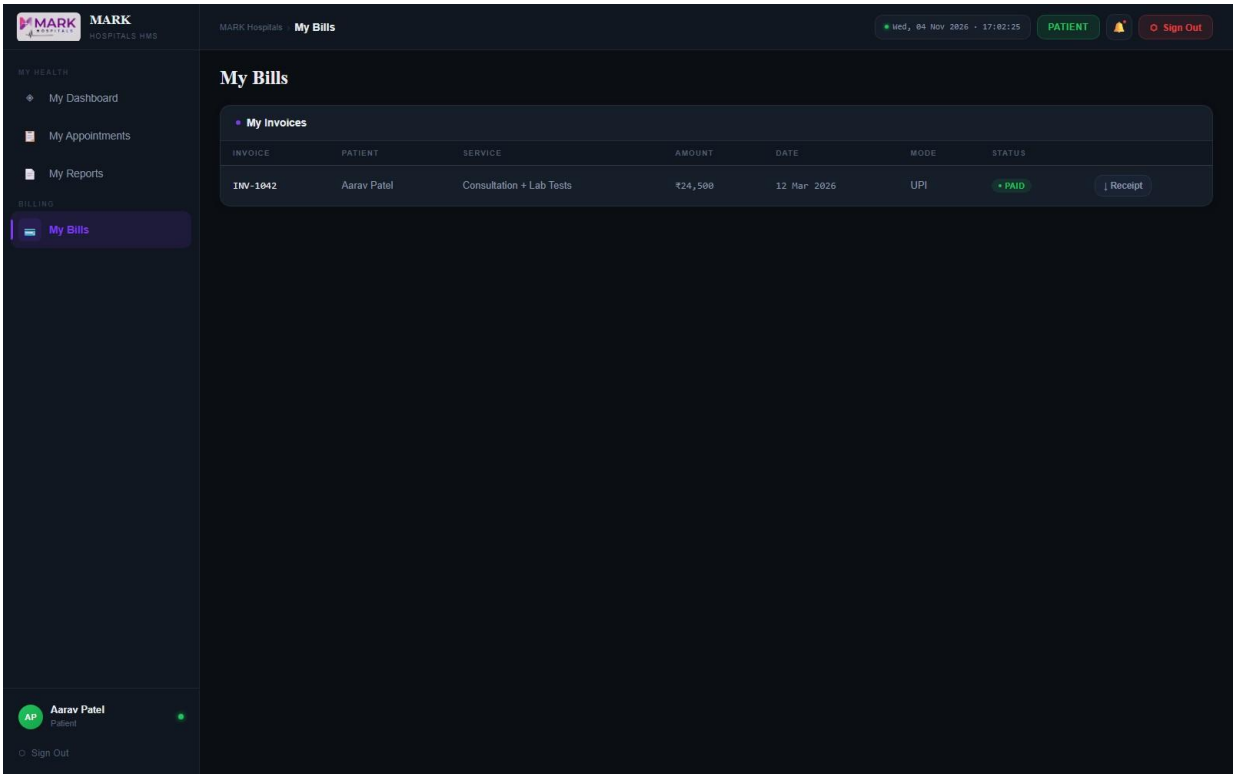
My Appointments



My Reports

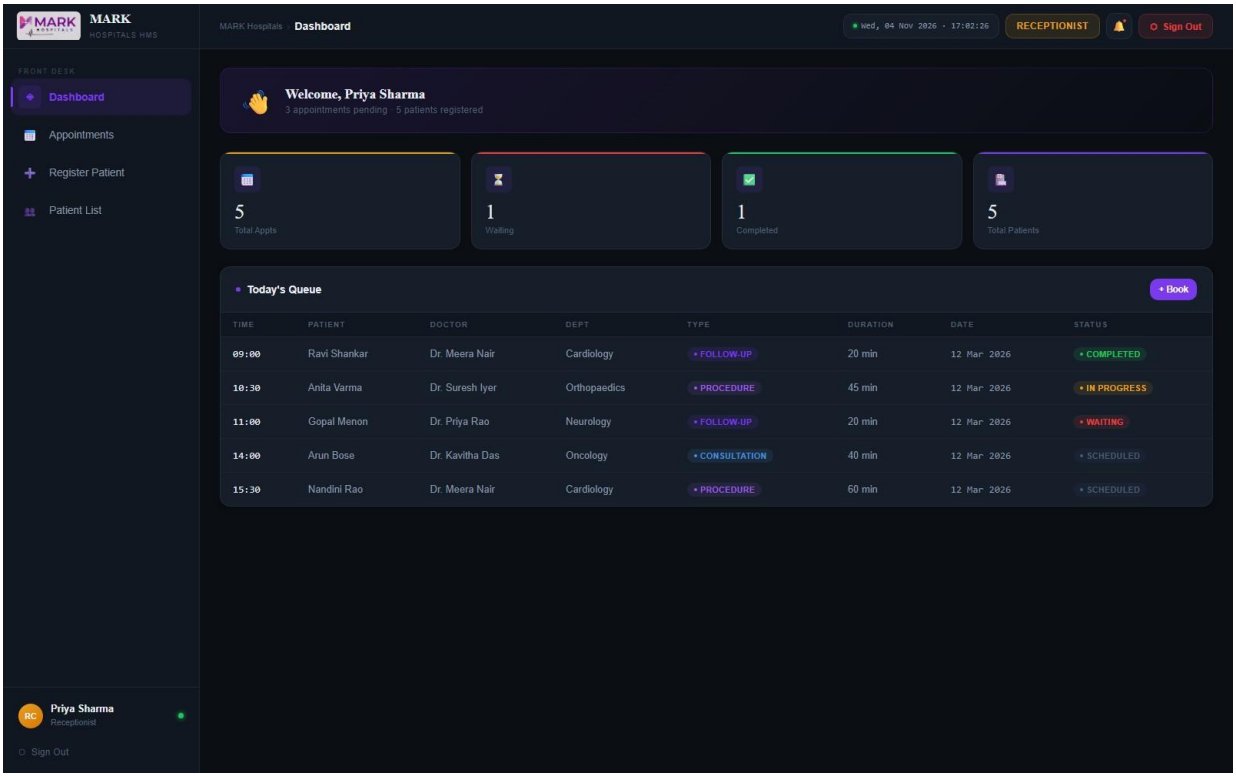

MARK
HOSPITALS HMS
MARK Hospitals
My Reports
Wed, 04 Nov 2026 · 17:02:25
PATIENT
Sign Out
MY HEALTH
My Dashboard
My Appointments
My Reports
BILLING
My Bills
My Reports
Lab Reports
TEST
RESULT
NORMAL RANGE
DATE
FLAG
HbA1c
7.2%
4.0–5.6%
10 Mar 2026
UNDER OBS.
↓ PDF
Blood Pressure
138/88 mmHg
120/80 mmHg
12 Mar 2026
UNDER OBS.
↓ PDF
AP
Aarav Patel
Patient
Sign Out

My Bills

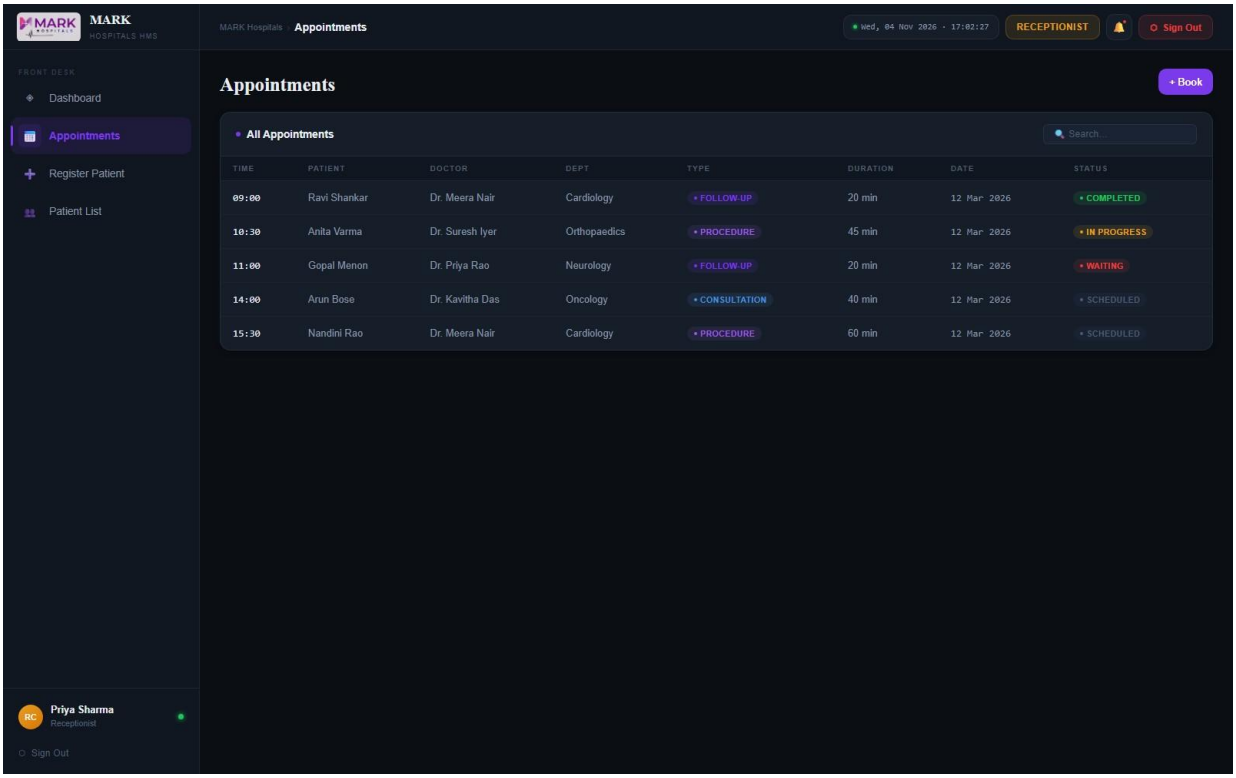

MARK
HOSPITALS HMS
MARK Hospitals
My Bills
Wed, 04 Nov 2026 · 17:02:25
PATIENT
Sign Out
MY HEALTH
My Dashboard
My Appointments
My Reports
BILLING
My Bills
My Bills
My Invoices
INVOICE
PATIENT
SERVICE
AMOUNT
DATE
MODE
STATUS
INV-1042
Aarav Patel
Consultation + Lab Tests
₹24,500
12 Mar 2026
UPI
PAID
↓ Receipt
AP
Aarav Patel
Patient
Sign Out

Receptionist

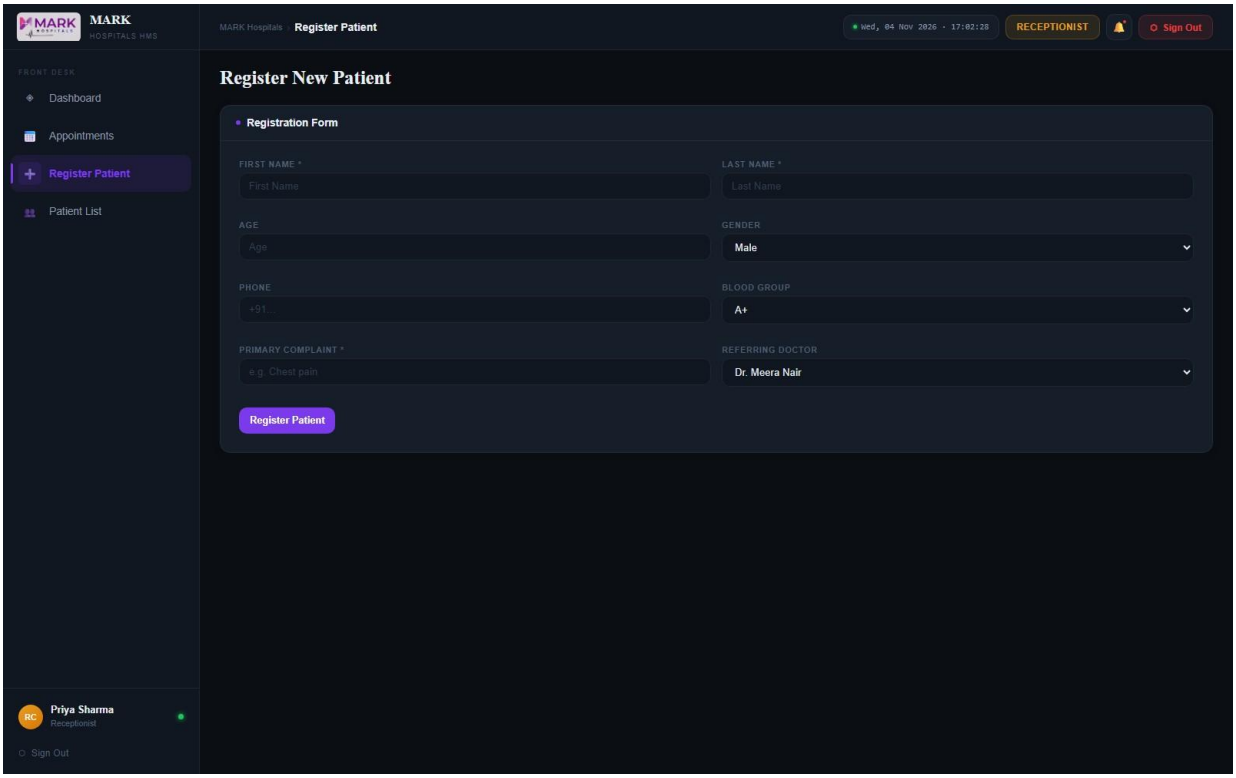
Dashboard



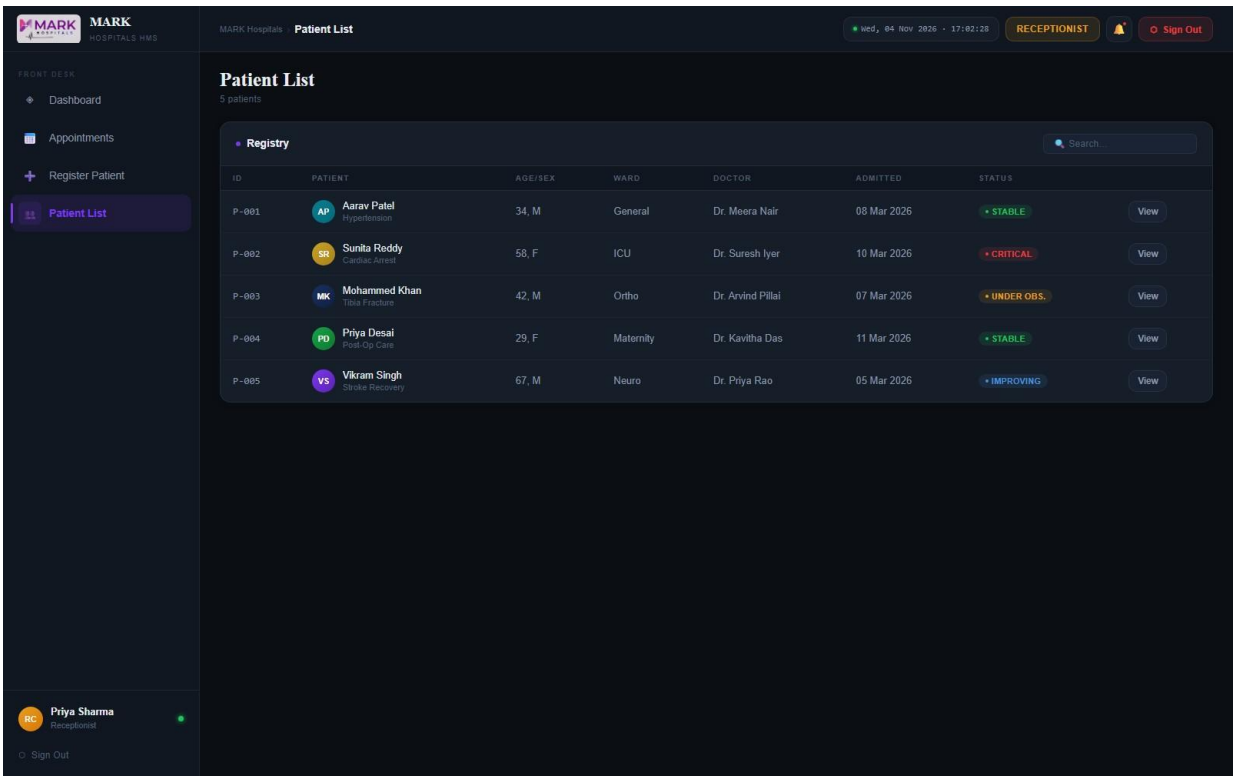
Appointments



Register Patient

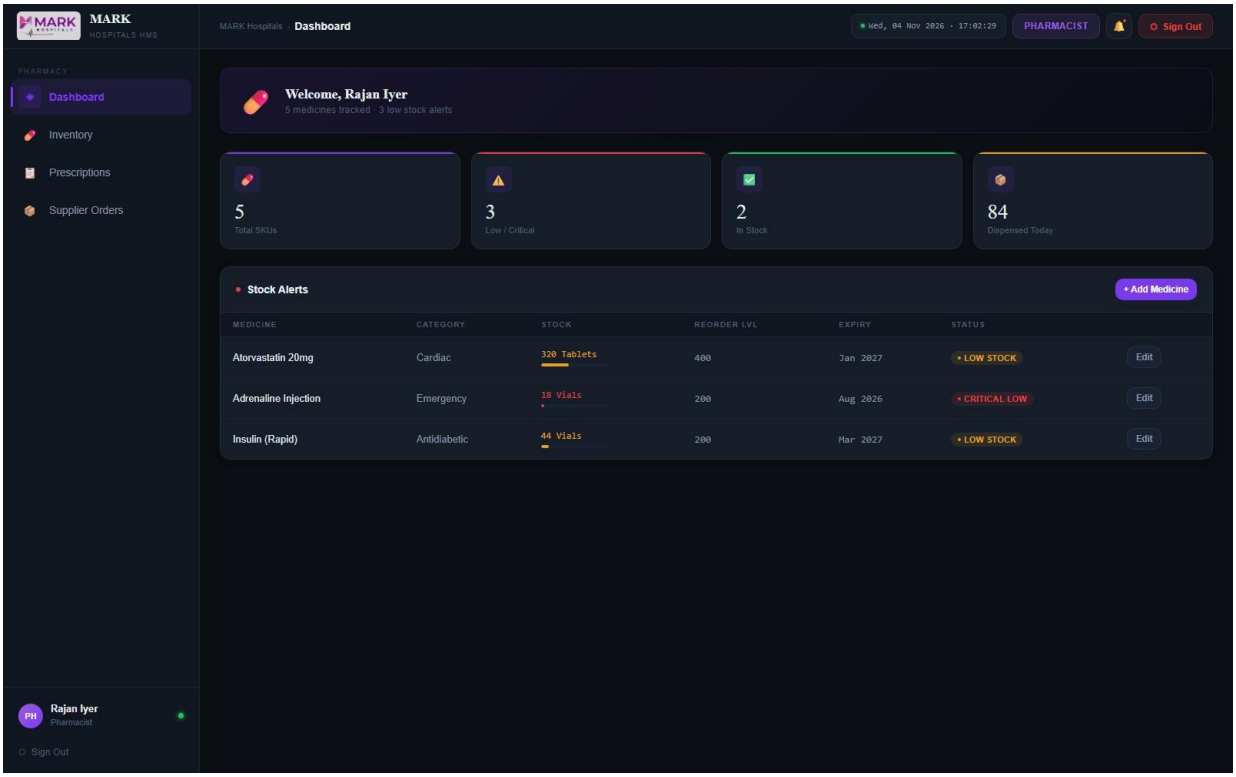


Patient List

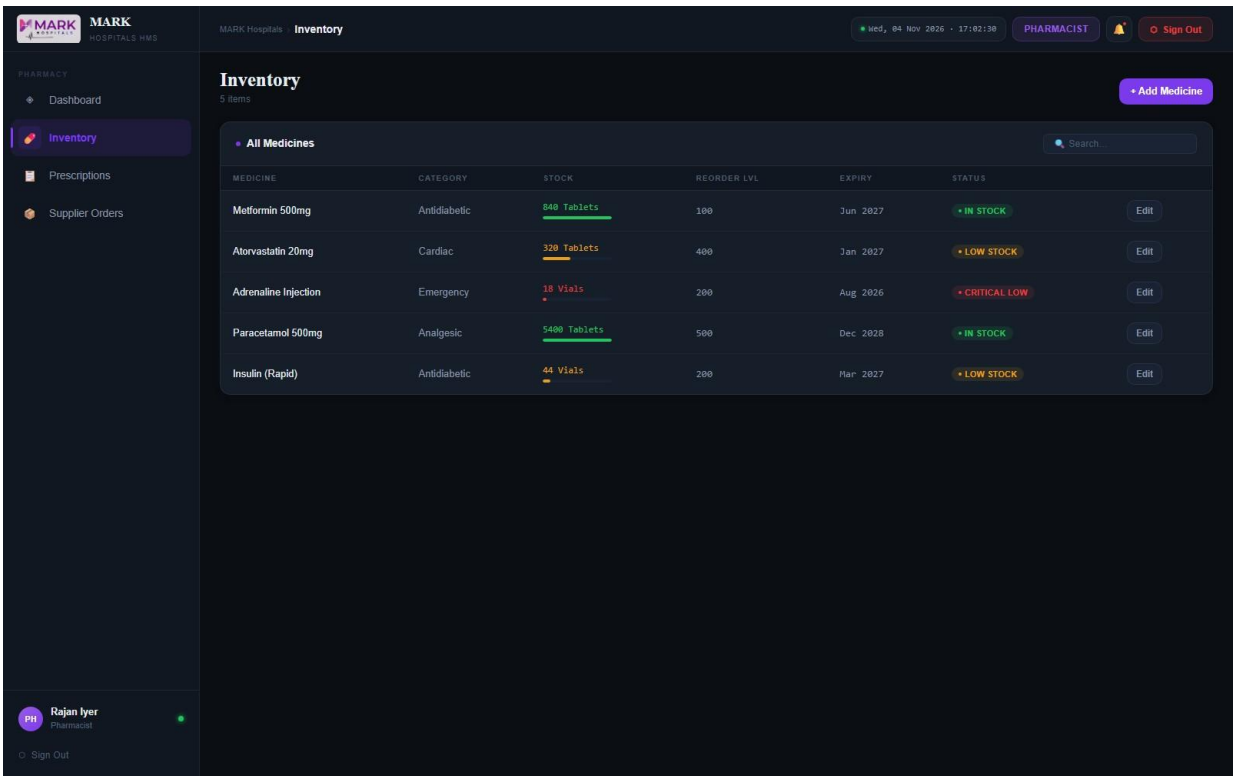


Pharmacist

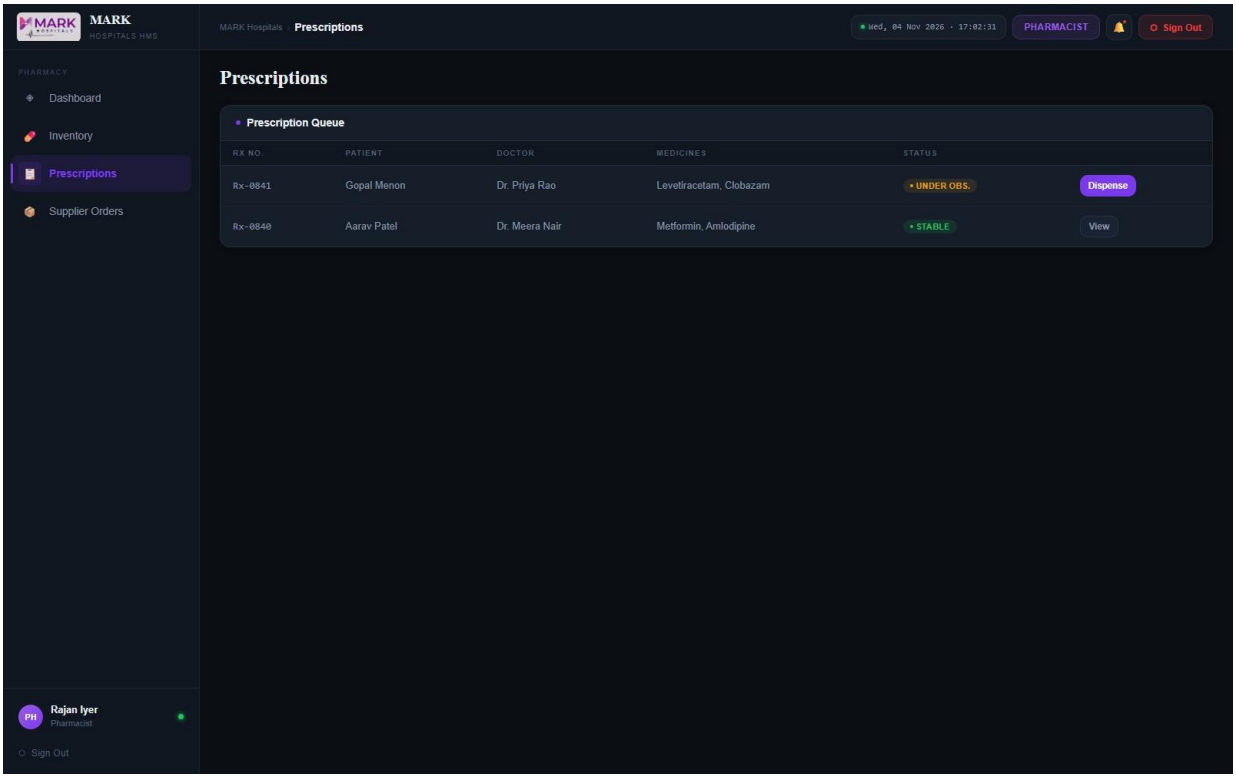
Dashboard



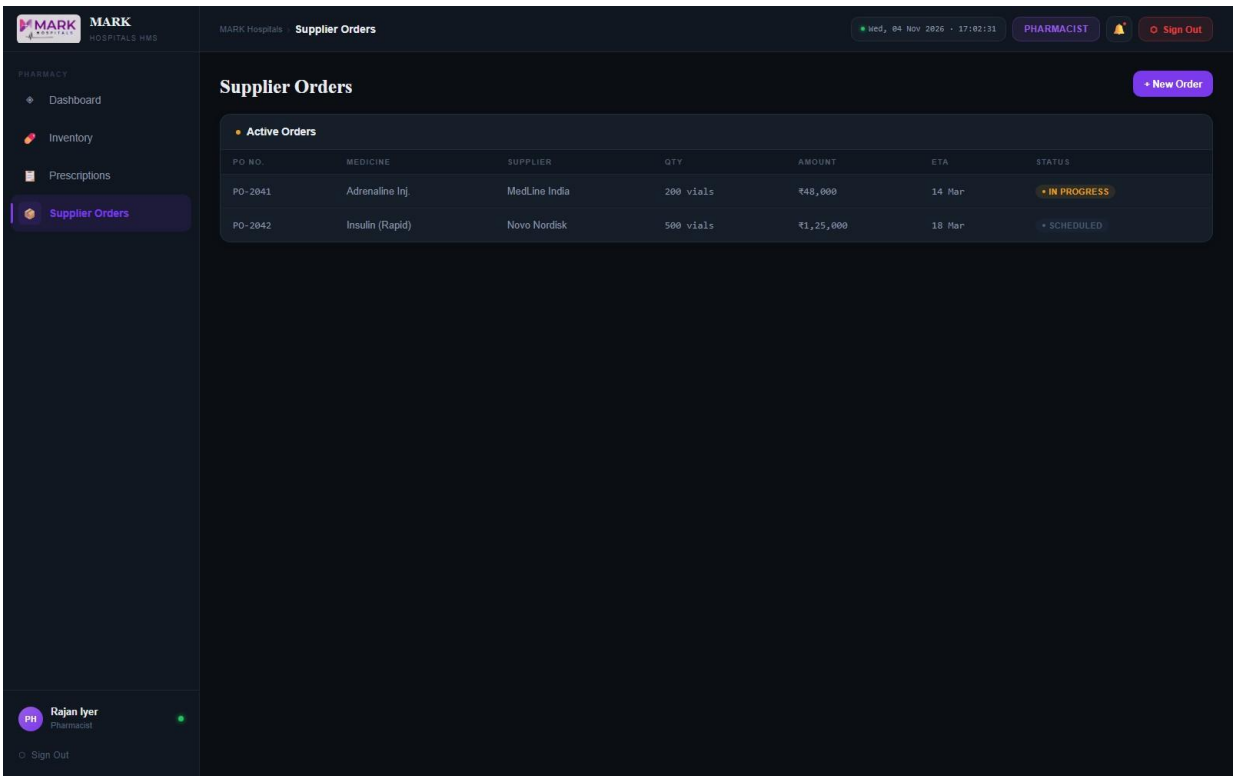
Inventory


MARK
HOSPITALS HMS
MARK Hospitals
Inventory
Wed, 04 Nov 2026 · 17:02:30
PHARMACIST
Sign Out
PHARMACY
Dashboard
Inventory
Prescriptions
Supplier Orders
Inventory
5 items
+ Add Medicine
All Medicines
Search...
MEDICINE
CATEGORY
STOCK
REORDER LVL
EXPIRY
STATUS
Metformin 500mg
Antidiabetic
840 Tablets
100
Jun 2027
IN STOCK
Edit
Atorvastatin 20mg
Cardiac
320 Tablets
400
Jan 2027
LOW STOCK
Edit
Adrenaline Injection
Emergency
18 Vials
200
Aug 2026
CRITICAL LOW
Edit
Paracetamol 500mg
Analgesic
5400 Tablets
500
Dec 2028
IN STOCK
Edit
Insulin (Rapid)
Antidiabetic
44 Vials
200
Mar 2027
LOW STOCK
Edit
PH
Rajan Iyer
Pharmacist
Sign Out

Prescriptions

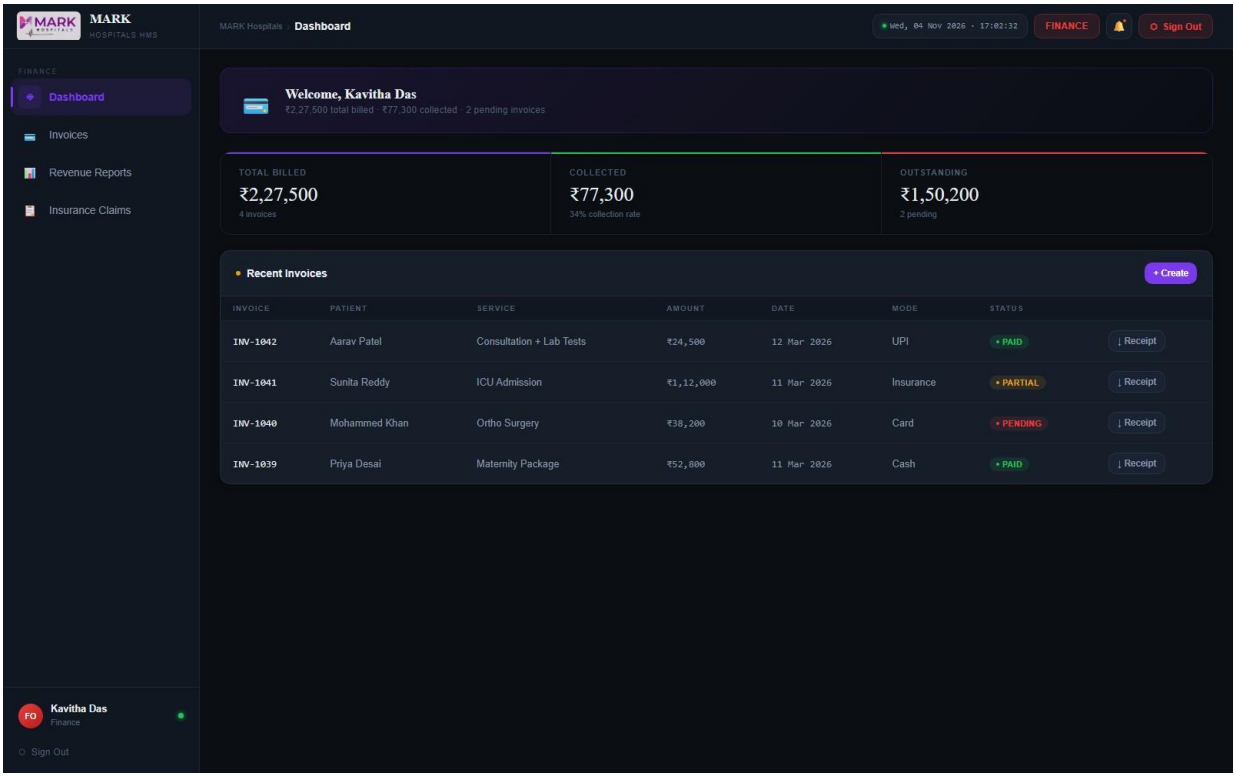


Supplier Orders

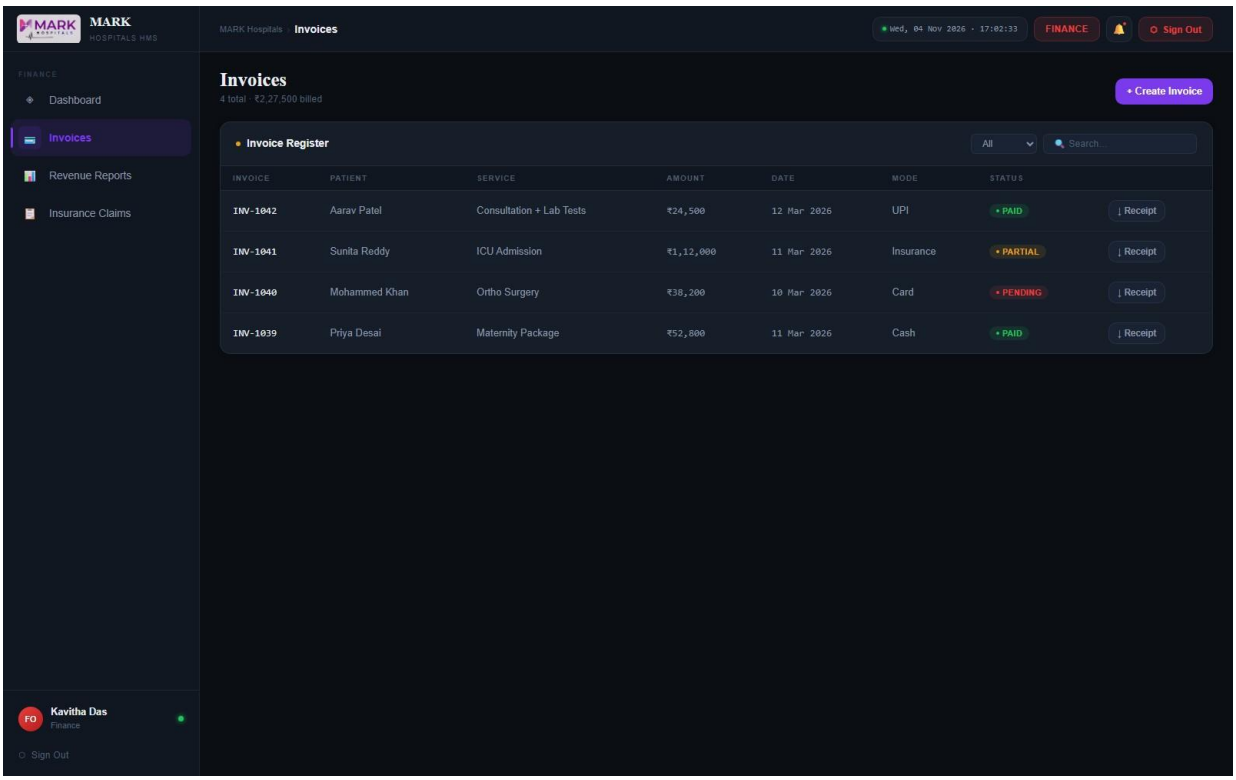


Finance

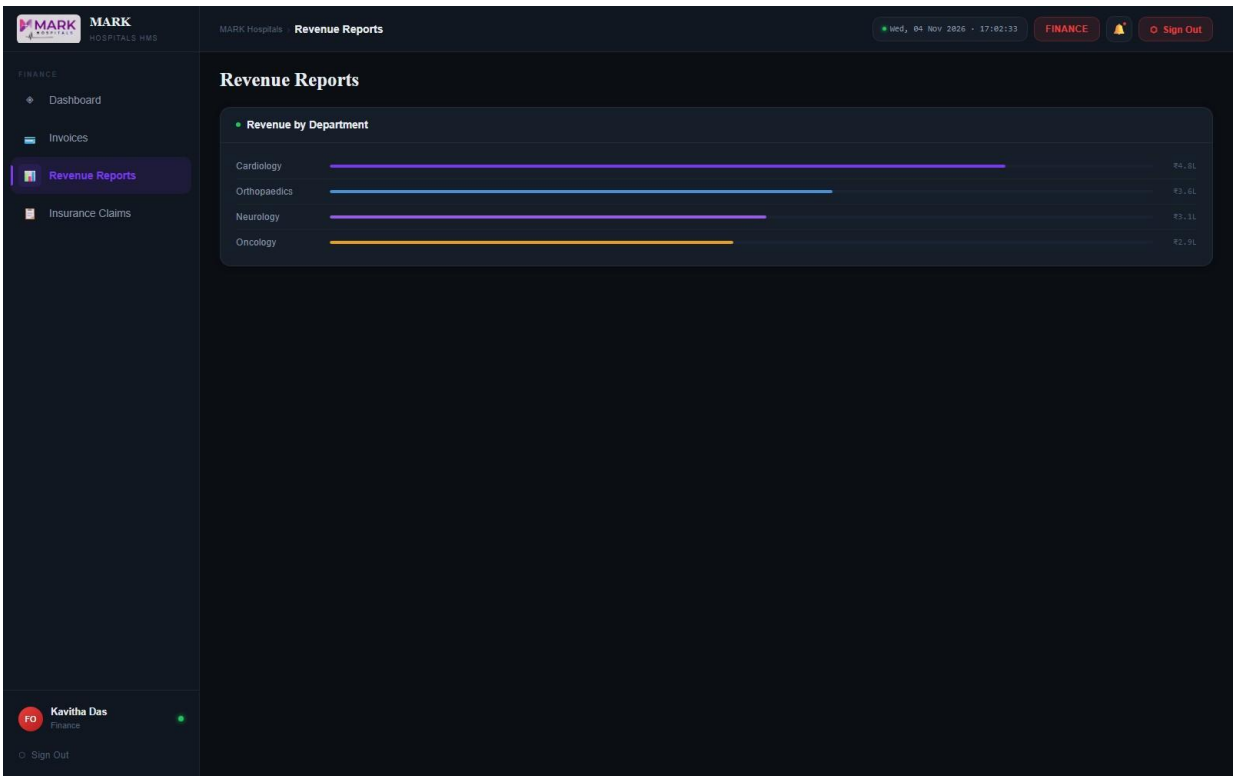
Dashboard



Invoices


MARK
HOSPITALS HMS
MARK Hospitals
Invoices
Wed, 04 Nov 2026 · 17:02:33
FINANCE
Sign Out
FINANCE
Dashboard
Invoices
Revenue Reports
Insurance Claims
Invoices
4 total · ₹2,27,500 billed
+ Create Invoice
Invoice Register
All
Search...
INVOICE
PATIENT
SERVICE
AMOUNT
DATE
MODE
STATUS
INV-1042
Aarav Patel
Consultation + Lab Tests
₹24,500
12 Mar 2026
UPI
PAID
↓ Receipt
INV-1041
Sunita Reddy
ICU Admission
₹1,12,000
11 Mar 2026
Insurance
PARTIAL
↓ Receipt
INV-1040
Mohammed Khan
Ortho Surgery
₹38,200
10 Mar 2026
Card
PENDING
↓ Receipt
INV-1039
Priya Desai
Maternity Package
₹52,800
11 Mar 2026
Cash
PAID
↓ Receipt
FO
Kavitha Das
Finance
Sign Out

Revenue Reports



Insurance Claims

